

Instructions: Answer all questions. Show your work for each problem by following the four problem-solving steps below.

Problem-Solving Steps

1. What am I given? What am I finding?
2. What equation connects them?
3. Plug in and solve.
4. Check the units and direction (if applicable).

Kinematics Formulas

$$\bar{v} = \frac{\Delta x}{t} \quad s = \frac{d}{t} \quad v = v_0 + at \quad \Delta x = v_0 t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a\Delta x$$

Hints

- "at rest" means the starting velocity is zero.
- Write the direction (north, south, east, west) for vector quantities.
- If a vector has no given direction, use positive (+) or negative (-).

1. A runner jogs 400 m in 100 s, then walks 200 m in 50 s. What is her average speed for the whole trip?

2. A train traveling at $10 \frac{\text{m}}{\text{s}}$ accelerates at a constant $3 \frac{\text{m}}{\text{s}^2}$ for 4 s. What is its final velocity?

3. A train starting from rest accelerates at a constant $2 \frac{\text{m}}{\text{s}^2}$ for 8 s. How far does it travel?

4. A car speeds up from $5 \frac{\text{m}}{\text{s}}$ to $25 \frac{\text{m}}{\text{s}}$ with a constant acceleration of $4 \frac{\text{m}}{\text{s}^2}$. How long does it take to reach $25 \frac{\text{m}}{\text{s}}$?

5. A car traveling at $8 \frac{\text{m}}{\text{s}}$ accelerates at a constant $3 \frac{\text{m}}{\text{s}^2}$. How long does it take to reach $20 \frac{\text{m}}{\text{s}}$?
6. A car speeds up from $10 \frac{\text{m}}{\text{s}}$ to $20 \frac{\text{m}}{\text{s}}$ over a distance of 30 m. What is its acceleration?
7. A car moving at $25 \frac{\text{m}}{\text{s}}$ brakes with a constant acceleration of $-5 \frac{\text{m}}{\text{s}^2}$. How far does it travel before it stops?
8. A stone is dropped from rest from the top of a tall building and falls for 4 s. ($g = 9.8 \frac{\text{m}}{\text{s}^2}$)
- (a) How far does the stone fall?
- (b) What is the stone's velocity just before it lands?